

Examiner  
only

3. Rhys and Elliot have a small piece of an unknown metal. The metal has an **irregular shape**. To identify the metal they find its density and compare the value to known values of the density of common metals.

(a) Describe a method they could use to find the density of the metal. [6 QER]

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(b) The table below gives data on the density of some common metals.

| Metal     | Density<br>(g/cm <sup>3</sup> ) |
|-----------|---------------------------------|
| aluminium | 2.70                            |
| copper    | 8.96                            |
| gold      | 19.32                           |
| iron      | 7.87                            |
| tin       | 7.26                            |

(i) Rhys and Elliot calculate the density of their metal to be 8.1 g/cm<sup>3</sup>.  
State which metal the irregular shape is most likely to be. Give a reason for your  
answer. [2]

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(ii) Rhys and Elliot are not confident that they can correctly identify the metal. Suggest  
why they think this. [1]

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(iii) The table below shows their results.

| Mass (g) | Volume (cm <sup>3</sup> ) | Density (g/cm <sup>3</sup> ) |
|----------|---------------------------|------------------------------|
| 65       | 8                         | 8.1                          |

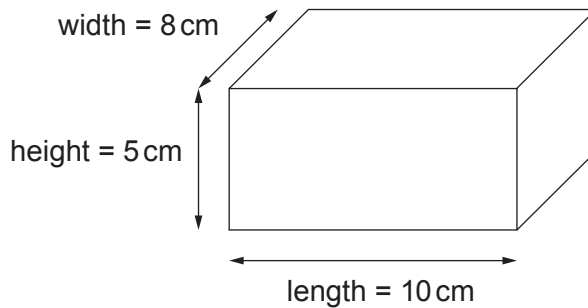
Suggest how Rhys and Elliot could get a more accurate value for the density. [1]

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- (c) The boys notice that gold has a high density of  $19.32 \text{ g/cm}^3$  and they are interested in the mass of a gold block.



- (i) Use the equation:

$$\text{volume} = \text{length} \times \text{height} \times \text{width}$$

to calculate the volume of the gold block shown above.

[1]

Volume = .....  $\text{cm}^3$

- (ii) Use the equation:

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the gold block.

[2]

Mass = ..... g

